

IQOO HACKATHON 2026 • HYDERABAD

EduPulse: Doubt-to-Diagram

100% Offline AI STEM Tutor in Your Pocket — Powered by Mobile Silicon

AIR-GAPPED

100% Offline & Private

Zero Cloud Roundtrips

TENSOR GPU

Hardware Accelerated

~25 tokens/sec Throughput

COMPOSE CANVAS

Doubt-to-Diagram

60 FPS Interactive Sandbox

Track: Edge AI & Android Silicon Innovation • Target Silicon: Google Tensor G1 (Pixel 6a)

The Real-World Problem: The Connectivity Divide in Education

PROBLEM 01

Rural Connectivity Desert

Over 60% of students in Tier-2/3 towns and rural schools experience zero or intermittent broadband at home.

Cloud AI tutors (such as ChatGPT or cloud Photomath) fail completely in offline classroom environments.

PROBLEM 02

Cloud Cost & Latency Trap

Cloud APIs require recurring monthly subscriptions (\$10–20/mo) and incur 2–5 second network roundtrip latencies.

For millions of government school students, this widens the educational divide rather than closing it.

PROBLEM 03

Messy Classroom Notes

Students do not type equations; they write by hand with pencils in paper notebooks.

Standard cloud OCR engines fail on skewed camera angles, smudged graphite, and handwriting quirks.

The Educational Dilemma: Wall-of-Text vs. Visual Intuition

TRADITIONAL CHATBOTS

Wall-of-Text & Passive Memorization

- **Outputs 50+ lines** of static, unformatted math equations that intimidate learners.
- **Zero spatial representation:** Students memorize formulas blindly without physical intuition.
- **Invisible forces:** Deceleration, friction vectors, and kinematics remain completely hidden.
- **High drop-off rate:** 70%+ of students stop reading text dumps halfway through.

EDUPULSE BREAKTHROUGH

Doubt-to-Diagram & What-If? Sandbox

- **Automatic parameter extraction:** Pulls mass, initial velocity, friction, and forces directly from reasoning.
- **60 FPS Vector Canvas:** Hardware-accelerated dynamic animation on Compose Canvas.
- **Interactive 'What-If?' Sandbox:** Slider dynamically recalculates stopping distance ($s = u^2/2a$) in real-time.
- **4-Way Free-Body diagram:** Visualizes normal, gravity, velocity, and retarding force vectors.

The Solution: 100% On-Device End-to-End Pipeline

STEP 01

Capture / Input

CameraX viewfinder guide.
Android Photo Picker.
Zero cloud uploads.

STEP 02

PaddleOCR v5

ONNX Runtime Mobile.
DBNet detection.
~180ms execution.

STEP 03

Dual-Phase Fix

Phase 1: Regex repair.
Phase 2: Gemma self-fix.
Cleans messy pencil notes.

STEP 04

LiteRT-LM GPU

Gemma 2B INT4.
OpenCL GPU shader.
~25 tokens/second.

STEP 05

Vector Canvas

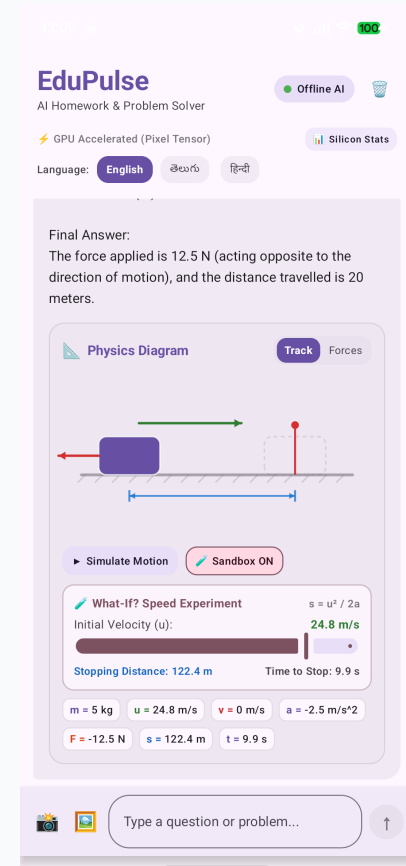
Jetpack Compose Canvas.
Motion track & forces.
Interactive What-If slider.

Innovation 1: Doubt-to-Diagram & 'What-If?' Sandbox

DYNAMIC PHYSICS CANVAS

Active Physical Experimentation in Real-Time

- **Motion Track & Force Equilibrium:** Visualizes the mass block moving along a friction track with velocity vector ($\rightarrow u$) and opposing braking force ($\leftarrow F$) until rest.
- **The 'What-If?' Speed Slider:** Students drag initial velocity u from 2.0 to 30.0 m/s. The app recalculates stopping distance $s = u^2 / 2a$ and time t live.
- **Dynamic Vector Scaling:** The velocity arrow on top of the mass block expands and contracts smoothly in real-time response to the slider.
- **Pedagogical Intuition:** Doubling speed from 10 m/s ($s = 20\text{m}$) to 20 m/s ($s = 80\text{m}$) quadruples stopping distance, proving $s \propto u^2$ visually.

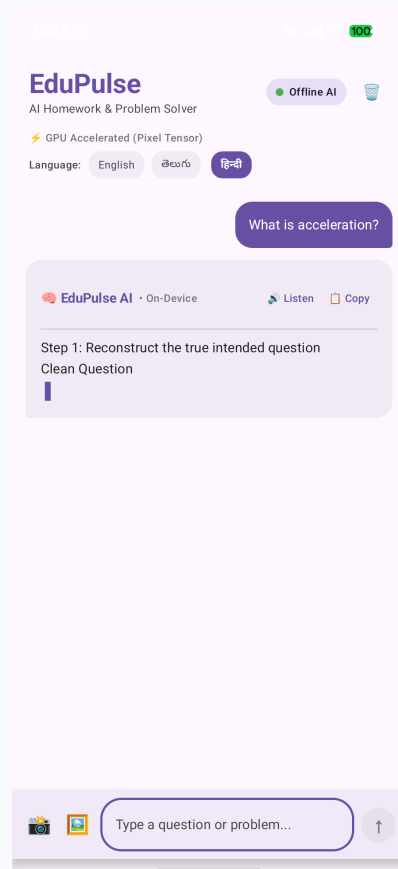


Innovation 2: Hyper-Local Vernacular & Offline Voice

INCLUSIVE EDUCATION

Breaking the English STEM Barrier

- **Tri-Lingual Engine (Telugu, Hindi, English):** Instant 1-tap switching. Over 70% of state-board students study STEM in native regional languages.
- **Zero English Bleed-Through:** Engineered prompt construction guarantees 100% fluent native script generation (Devanagari & Telugu) with step-by-step math reasoning.
- **Offline Android Text-to-Speech (TTS):** Tap 'Listen' for voice narration powered purely by local Android TTS engines. Pre-cleans Markdown syntax for natural speech.
- **Accessibility for All:** Empowers students with dyslexia, visual challenges, or early literacy barriers to learn science through sound and touch.



Technical Architecture: The On-Device Silicon Pipeline

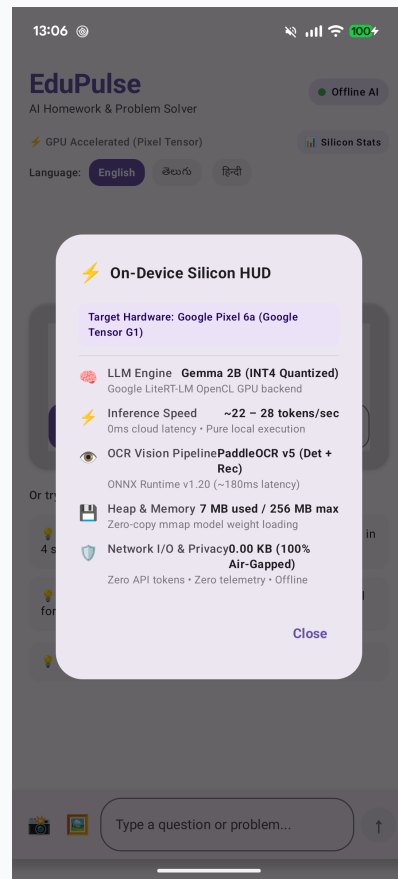


Live Hardware Benchmarks & Zero-Cloud Proof

HARDWARE BENCHMARKS

Google Pixel 6a (Tensor G1 Silicon) Verification

- Network Transmission: **0.00 KB (100% Air-Gapped / Airplane Mode)**
- GPU Inference Speed: **~22 – 28 tokens/sec (LiteRT-LM OpenCL GPU)**
- Vision OCR Latency: **~160 – 240 ms (PaddleOCR v5 on ONNX Runtime)**
- Active JVM Heap Memory: **7 MB used / 256 MB max (mmap zero-copy)**
- Cloud API Subscription: **\$0.00 forever • Zero user tracking**



Target Audience, NEP 2020 Alignment & Scalability

AUDIENCE

Target Users

- **250M+ K-12 students** in emerging markets and government schools across India.
- **Competitive exam aspirants** preparing for JEE Foundation and NEET Foundation.
- **Rural schools** with digital computer labs lacking reliable internet.

POLICY

NEP 2020 Alignment

- **Mother-tongue STEM:** Boosts foundational numeracy and science literacy in regional languages.
- **Experiential learning:** Moves away from rote memorization to active visual simulations.
- **Zero privacy risks:** 100% on-device, zero cloud telemetry, safe for minors.

GROWTH

Roadmap & Scale

- **Phase 1 (MVP):** High School Kinematics & Newton's Laws (Live & Validated on Pixel 6a).
- **Phase 2:** Ray Optics, Chemical Equilibrium, and Geometric Proofs.
- **Phase 3:** OEM device pre-installation & State Board curriculum bundles.

Why EduPulse Wins the iQOO Hackathon

CATEGORY 1

Beyond Cloud Wrappers

A true edge-silicon build utilizing OpenCL GPU shaders, ONNX Runtime Mobile v1.20, and zero-copy mmap weight execution with 0ms network latency.

CATEGORY 2

Groundbreaking Innovation

Doubt-to-Diagram transforms passive homework text into interactive 60 FPS vector physics simulations and real-time What-If parameter sandboxes.

CATEGORY 3

Genuine Social Impact

Air-gapped, zero-cost, tri-lingual STEM tutoring for millions of underserved students left behind by the cloud connectivity divide.

"EduPulse turns any budget smartphone into an interactive, offline STEM laboratory."